

Refraction and Snell's Law - HKDSE Physics Summary

1. Refraction of Light

1.1 Definition

Refraction is the bending of light when it passes from one medium to another with different optical density.

Key Points:

- Light changes **speed** when entering different medium
- Light changes **direction** (except at normal incidence)
- Frequency remains constant
- Wavelength changes

1.2 Key Terms

- **Incident ray** (入射光線) : Light ray approaching the boundary
 - **Refracted ray** (折射光線) : Light ray after crossing boundary
 - **Normal** (法線) : Perpendicular line to boundary at point of incidence
 - **Angle of incidence** (入射角) : Angle between incident ray and normal (i)
 - **Angle of refraction** (折射角) : Angle between refracted ray and normal (r)
 - **Optical density** (光密度) : Measure of how much light slows in medium
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2. Behavior of Refraction

2.1 Light Entering Denser Medium

When light travels from **less dense to more dense** medium (e.g., air $\rightarrow$ water, air $\rightarrow$ glass):

- Light **slows down**
- Light **bends towards normal**

- Angle of refraction < Angle of incidence (
 $r < i$
)

Examples:

- Air → Water
- Air → Glass
- Water → Glass

2.2 Light Entering Less Dense Medium

When light travels from **more dense to less dense** medium (e.g., water → air, glass → air):

- Light **speeds up**
- Light **bends away from normal**
- Angle of refraction > Angle of incidence (
 $r > i$
)

Examples:

- Water → Air
- Glass → Air
- Glass → Water

2.3 No Refraction Cases

Light does **not** bend when:

1. **Normal incidence:** Light hits boundary at 90° (along normal)
 2. **Same medium:** Light stays in same medium (no boundary crossed)
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3. Refractive Index

3.1 Definition

Refractive index (n) measures how much light slows down in a medium compared to vacuum.

$$n = \frac{c}{v}$$

Where:

- n
= refractive index (no unit)
- c
= speed of light in vacuum (3×10^8 m s⁻¹)
- v
= speed of light in the medium

3.2 Common Refractive Indices

Medium	Refractive Index (n)
Vacuum	1.00 (exactly)
Air	1.00 (≈ 1.0003)
Water	1.33
Glass	1.50 (typical)
Diamond	2.42

Important:

- Higher n
→ Light travels slower → Optically denser
 - Lower n
→ Light travels faster → Optically less dense
 - $n \geq 1$
always (vacuum is minimum)
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4. Snell's Law

4.1 The Law

Snell's Law relates angles of incidence and refraction to refractive indices:

$$n_1 \sin i = n_2 \sin r$$

Where:

- n_1
= refractive index of first medium
- i
= angle of incidence
- n_2
= refractive index of second medium
- r
= angle of refraction

4.2 Alternative Forms

When light enters from air (

$$n_1 \approx 1$$

):

$$\sin i = n \sin r$$

or

$$n = \frac{\sin i}{\sin r}$$

Relationship between speeds:

$$\frac{n_1}{n_2} = \frac{v_2}{v_1}$$

5. Real and Apparent Depth

5.1 Apparent Depth Phenomenon

When viewing an object underwater from above:

- Object appears **closer to surface** than it actually is

- Object appears **shallower**
- Due to refraction at water-air boundary

5.2 Real vs Apparent Depth Formula

$$n = \frac{\text{Real depth}}{\text{Apparent depth}}$$

or

$$\text{Apparent depth} = \frac{\text{Real depth}}{n}$$

For water (

$$n = 1.33$$

):

- Object at real depth 4 m appears at 3 m
- Apparent depth $\approx 0.75 \times$ Real depth

5.3 Applications

- Swimming pool appears shallower than actual depth
 - Fish appears closer to surface than it really is
 - Spear fishing requires aiming below apparent position
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6. Dispersion of Light

6.1 Definition

Dispersion is the separation of white light into its component colors due to different refractive indices for different wavelengths.

6.2 Mechanism

- Different colors have different wavelengths
- Different wavelengths refract by different amounts
- Shorter wavelength $\rightarrow$ Higher refractive index $\rightarrow$ Bends more
- Longer wavelength $\rightarrow$ Lower refractive index $\rightarrow$ Bends less

6.3 Color Order in Prism

When white light passes through a prism:

Order (least to most deviated):

Red → Orange → Yellow → Green → Blue → Indigo → Violet

Refractive index order:

$$n_{\text{red}} < n_{\text{orange}} < n_{\text{yellow}} < n_{\text{green}} < n_{\text{blue}} < n_{\text{indigo}} < n_{\text{violet}}$$

Mnemonic: ROY G BIV

6.4 Applications

- **Rainbow:** Dispersion and reflection in water droplets
 - **Prism:** Separates white light into spectrum
 - **Chromatic aberration:** Unwanted color separation in lenses
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7. Applications of Refraction

7.1 Optical Fibers

- Uses total internal reflection (special case of refraction)
- Transmits light signals over long distances
- Used in telecommunications, internet, medical endoscopes

7.2 Lenses

- Convex and concave lenses use refraction
- Camera, eyeglasses, microscope, telescope
- Control light path by refraction at curved surfaces

7.3 Atmospheric Refraction

- **Sunrise/Sunset:** Sun visible before it rises/after it sets
- **Twinkling stars:** Light refracted by moving air layers
- **Mirages:** Hot air near ground causes refraction

7.4 Everyday Examples

- Straw appears bent in water
- Swimming pool appears shallower

- Fish appears at wrong position
 - Magnifying effect of water droplet
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8. Problem-Solving Strategies

8.1 Ray Diagram Method

1. Draw normal at boundary
2. Draw incident ray to boundary
3. Determine if entering denser or less dense medium
4. Bend ray towards normal (denser) or away from normal (less dense)
5. Measure angles from normal

8.2 Snell's Law Calculations

Steps:

1. Identify given values:

n_1

,

n_2

,

i

, or

r

2. Write Snell's Law:

$$n_1 \sin i = n_2 \sin r$$

3. Substitute known values
4. Solve for unknown
5. Check if answer is reasonable

Tips:

- Angles always measured from normal
- Use calculator in degree mode
- Check: Denser medium $\rightarrow$ smaller angle

8.3 Apparent Depth Calculations

Steps:

1. Identify real depth
2. Identify refractive index

n

3. Apply:

$$\text{Apparent depth} = \frac{\text{Real depth}}{n}$$

4. Or:

$$n = \frac{\text{Real depth}}{\text{Apparent depth}}$$

9. Common Misconceptions

9.1 Misconception 1

✗ **Wrong:** Light always bends when crossing boundary

✓ **Correct:** Light does not bend at normal incidence (along normal)

9.2 Misconception 2

✗ **Wrong:** Light bends towards normal when entering less dense medium

✓ **Correct:** Light bends **away from** normal when entering less dense medium

9.3 Misconception 3

✗ **Wrong:** Refractive index can be less than 1

✓ **Correct:** Refractive index

$$n \geq 1$$

always (vacuum is minimum with

$$n = 1$$

)

9.4 Misconception 4

✗ **Wrong:** Frequency changes during refraction

✓ **Correct:** Frequency remains constant; only speed and wavelength change

9.5 Misconception 5

✗ **Wrong:** Red light has higher refractive index than violet

✓ **Correct:** Violet has higher

n

than red; violet bends more

10. Important Relationships

10.1 Speed and Refractive Index

$$n = \frac{c}{v}$$

- Higher

n

→ Lower speed

- Lower

n

→ Higher speed

10.2 Wavelength Change

$$\frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1}$$

- Entering denser medium → Wavelength decreases
- Entering less dense medium → Wavelength increases

10.3 Frequency Constant

$$f_1 = f_2$$

- Frequency does **not** change during refraction
 - Determined by source, not medium
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11. HKDSE Exam Key Points

11.1 Must Remember

1. **Refraction:** Bending of light when changing medium
2. **Denser medium:** Light slows, bends towards normal
3. **Less dense medium:** Light speeds up, bends away from normal
4. **Refractive index:**

$$n = \frac{c}{v}$$

,

$$n \geq 1$$

5. **Snell's Law:**

$$n_1 \sin i = n_2 \sin r$$

6. **Apparent depth:**

$$n = \frac{\text{Real depth}}{\text{Apparent depth}}$$

7. **Dispersion:** Red bends least, violet bends most

11.2 Common Question Types

1. **Ray diagrams:** Draw refracted ray
2. **Snell's Law calculations:** Find angle or refractive index
3. **Apparent depth:** Calculate real or apparent depth
4. **Dispersion:** Explain color separation
5. **Applications:** Identify uses of refraction

11.3 Exam Tips

- Always measure angles from **normal**
- Check calculator is in **degree mode**
- Draw neat diagrams with ruler
- Label: Normal, incident ray, refracted ray, angles
- State clearly: Towards or away from normal
- Show all working in calculations

12. Quick Memory Mnemonics

Refraction Direction:

To denser, towards normal
To less dense, away from normal
Slow down, bend in
Speed up, bend out

Snell's Law:

$n_1 \sin i = n_2 \sin r$
First medium, first angle
Second medium, second angle

Apparent Depth:

n equals real over apparent
Looks shallower than it is

Dispersion Order:

ROY G BIV
Red bends least, violet most
Red has lowest n , violet highest

Refractive Index:

$n = c \text{ over } v$
Higher n , slower speed
Always greater than or equal to 1

Study Suggestions:

1. Practice drawing ray diagrams for different media

2. Memorize Snell's Law and refractive index formula
3. Understand relationship between density, speed, and bending
4. Practice calculations with different scenarios
5. Learn common refractive indices (air, water, glass)
6. Understand dispersion and color order